

PATENT APPLICATION**IN THE UNITED STATES PATENT AND TRADEMARK OFFICE**

In re application of:

Jens Steen OLSEN

Appln. No.: 12/595,715

COE

Confirmation No.: 5595

Filed: November 17, 2009

Docket No: Q115476

Group Art Unit: 1655

Examiner: HOFFMAN, SUSAN

For: EXTRACT OF TRIGONELLA FOENUM-GRAECUM

AMENDMENT UNDER 37 C.F.R. § 1.111

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Office Action dated April 13, 2012, please amend the above-identified application as follows on the accompanying pages.

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AMENDMENTS TO THE SPECIFICATION

Page 2, after the last paragraph, kindly insert the following:

BRIEF DESCRIPTION OF THE DRAWINGS

Figure 1 is a chromatogram relating to an Example provided in the specification.

AMENDMENTS TO THE CLAIMS

This listing of claims will replace all prior versions and listings of claims in the application:

LISTING OF CLAIMS:

1 to 35. (canceled)

36. (currently amended): An extract ~~obtainable~~obtained by a process comprising the steps of

(a) preparing a mixture of a ~~plant material~~ seeds obtained from Trigonella foenum-graecum and aqueous liquid,

(b) incubating said mixture for at least 3 hours at a temperature between 0 and 45°C,

(c) heating of said mixture for a period of time sufficient to release the embryo from the seeds, and

(d) recovering a liquid extract from mixture.

37. (previously presented): The extract according to claim 36, wherein said plant material is seeds of said plant.

38. (previously presented): The extract according to claim 36, wherein said heating of said mixture is boiling for at least 5 minutes.

39. (previously presented): The extract according to claim 36, wherein the liquid is aqueous.

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40. (currently amended): The extract according to claim 36, comprising the additional step of freezing the incubated mixture prior to the heat treatment according to step (c) or freezing the heat-treated mixture.

41. (currently amended): The extract according to claim 40, wherein said additional step comprises freezing the heat-treated mixture and repeating step (c).

42. (previously presented): An extract fraction obtainable by performing reverse phase chromatography on a size B Lichroprep RP-18 (40-63 μ m) (Merck) of the extract according to claim 36 using the following gradient: 0-1 min H₂O/AcN 98:2, then using a steady gradient from 1-40 min going to 100% AcN and collecting the fraction at the time interval between 5 and 10 min.

43. (previously presented): The extract according to claim 36, wherein the solvent partly or entirely is removed by a method selected among the group consisting of: membrane filtration, evaporation, precipitation, extraction, azeotrope distillation, lyophilisation, spray drying and combinations thereof.

44. (previously presented): A composition comprising the extract according to claim 36.

45. (previously presented): The composition according to claim 44, wherein the composition is a pharmaceutical composition.

46. (previously presented): The composition according to claim 44, which is a disinfection agent.

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47. (withdrawn): A method for the treatment or prevention of an inflammatory condition, wherein a person suffering from an inflammatory condition is administered an amount of a pharmaceutical composition comprising the extract according to claim 36 in an amount sufficient to cure or alleviate the condition.

48. (withdrawn): A method according to claim 47, wherein the inflammatory condition is selected from diaper dermatitis, acne, asthma, or an inflammatory condition in the gastro-intestinal tract.

49. (withdrawn): A method for the treatment or prevention of an infectious condition, wherein a person suffering from an infection is administered an amount of a pharmaceutical composition comprising the extract according to claim 36 in an amount sufficient to cure or alleviate the condition.

50. (withdrawn): A method according to claim 48, wherein said infectious condition is selected from the group consisting of an infected wound, a surgical wound, impetigo, a Staphylococcus, such as Staphylococcus Aureus infection, an infection in the mouth cavity, periodontal disease, an infection in the eye or the adnexa of the eye, common cold, cold sores, Herpes simplex, warts, and pharyngitis.

51. (withdrawn): The method according to claim 48, wherein said infectious condition is Herpes simplex.

52. (withdrawn): The method according to claim 48, wherein said periodontal disease is selected among gingivitis, periodontitis, and halitosis,

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53. (withdrawn): A method according to claim 48, wherein the pharmaceutical composition is formulated as a mouth-wash, chewing gum, tooth-paste, balm, plaster, lip salve, spray, ointment, gel, capsule, drop, or tablet.

54. (withdrawn): A method for disinfection, wherein a disinfection agent comprising an extract according to claim 36 is applied to skin; installations such kitchens, slaughter houses, toilets and stables; kitchen hardware such as refrigerators, freezers; water; water-containers; or concrete elements in an amount sufficient for reducing risk of infection.

55. (withdrawn): A method for preservation of a food product, wherein the food product is subjected to treatment with a preservation agent comprising an extract according to claim 36 in an amount sufficient for retarding degradation of the food product.

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REMARKS**Preliminary Matters:**

This Application claims the benefit of priority from U.S. Provisional Application Serial No. 60/911,653, filed April 13, 2007. Applicant respectfully requests that the Examiner acknowledge Applicant's claim to priority and receipt of the priority document.

Specification:

The Examiner has objected to the specification for lacking the section "Brief Description of the Drawings". Applicant has amended the specification to overcome this objection.

Disposition of Claims:

Claims 36-55 are all the claims pending in the application. Of these claims 36-46 are rejected and claims 47-55 have been withdrawn from further consideration.

Claim Rejections Under 35 U.S.C. §112:

The Examiner has rejected claims 36-46 under 35 U.S.C. § 112 (first paragraph) as failing to comply with the written description requirement. Further, claim 43 is rejected under 35 U.S.C. § 112 (second paragraph) as being indefinite for failing to particularly point out and distinctly claim the subject invention. It is submitted that the above amendment overcomes this rejection.

Claim Rejections Under 35 U.S.C. § 102 and 103:

Claim 1 has been amended above. The basis for the amendment may be found in claim 3 (seeds), claim 9 (aqueous liquid), p. 3, 1. 35 (temperature), and p. 4, 1. 18 (embryo release).

Novelty

The Examiner has cited 4 documents against the novelty of the present invention. Hardman (US 3,620,919) discloses a process for recovering diosgenin from e.g. Trigonella by incubating the vegetable material in an aqueous medium for up to 72 hours,

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hydrolysing the incubated product with hydrochloric acid and solvent extracting sapogenins from the hydrolysate.

Hardman does not use the acidic aqueous extract from the process. It appears that the acidic extract is discarded and instead the acid-insoluble residue is extracted with petroleum ether. The petroleum ether phase is subsequently evaporated to leave the crude diosgenin, which is the active compound.

Lee (US 2005/0238728) discloses a method for preparing an extract. In a first step the Fenugreek seeds are soaked in water for a specified amount of time. After the soaking the seeds are crushed to separate the seeds in an endosperm and a testa fraction. Subsequently the endosperm and testa fractions are subjected to one or two extraction steps. In the first extraction step an organic solvent selected among hexane, cyclohexane, or ether is used, see [0098]. During the extraction the mixture of organic solvent and seed fractions may be heated one or more times. If a second extraction step is applied, the solvent is ethanol or a solvent with similar properties, see [0101].

Thus, Lee does not disclose the heating of a mixture of seeds and aqueous liquid.

Ziya Al-Din et al. discloses an ancient therapeutic composition comprising trigonella foenum-graecum Linn. In a first step a powder of sicklefruit fenugreek is soaked overnight in water and then boiled till water remains half. Subsequently sugar is added and the mixture is boiled to desired consistency.

Ziya Al-Din et al. does not disclose the use of whole seeds. Only a powder of the seeds are used.

Mohammad Akmal et al. discloses a pharmaceutical composition comprising 7 different ingredients: cultivated licorice, wild celery, sweet fennel, jwarancusa grass, wine grape, common fig, and sicklefruit fenugreek. The ingredients are blended together and soaked in water overnight. Subsequently the mixture is boiled till 2/3 of water remains. The citation does not mention the temperature of the soaking step and it is not indicated whether the boiling is sufficient for the endosperm to be released.

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Non-obviousness

The examiner rejects the application based in §102/103 in combination with MPEP § 2113.

Essential for the present invention is the incubation step, where seeds mixed with water are incubated for a certain time so that the sprouting initiates. Typically, the incubation step takes 24 hours, see e.g. examples 1 and 14 of the description. In the incubation step the seeds have swelled and the weight increased but a sprout is usually not visible when the heat treatment is commenced. As it is well known from brewing, the sprouting process generates chemical substances which are not otherwise present or not present in the same amount. While it is not wished to be bound by a certain theory it is presently believed that the compounds generated during the sprouting process eventually appear in the extract. In the prior art processes not including an incubation step as disclosed in the present application the special compounds produced during sprouting is not present at all or not present in sufficient amounts in the final extract.

Hardman uses the non-aqueous fraction, which is produced by extraction of the acidinsoluble residue with petroleum ether. The compounds extractable using petroleum ether are not the same as the compounds produced according to the present invention, which are extractable by an aqueous liquid, such as water.

The same applies for Lee, because the active ingredients are extracted with organic solvents — and not an aqueous liquid as required in the present invention.

The Examiner combines Hardman and Lee to remove the non-obviousness of claim 40 and 41. Even though Lee discloses a freeze-drying step, Lee does not disclose the extraction with an aqueous liquid. Lee uses organic solvents as stated above. Therefore, the person skilled in the art is not able to arrive at the present invention.

In an attempt to remove the non-obviousness of claims 42, the Examiner combined Hardman with Yang. Yang discloses an HPLC method for determining the contents of diosgen. However, Yang does not disclose a method for aqueous extraction. Therefore, the person skilled in the art will not be in a position to arrive within the frames of the present claims. Thus, it is submitted that the claims are patentable over the prior art.

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Conclusion:

In view of the above, reconsideration and allowance of this application are now believed to be in order, and such actions are hereby solicited. If any points remain in issue which the Examiner feels may be best resolved through a personal or telephone interview, the Examiner is kindly requested to contact the undersigned at the telephone number listed below. The USPTO is directed and authorized to charge all required fees, except for the Issue Fee and the Publication Fee, to Deposit Account No. 19-4880. Please also credit any overpayments to said Deposit Account.

Respectfully submitted,

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